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Elliot Yarnell Davies

Employment

2025–2028 **Massachusetts Institute of Technology**, *MIT Kavli Postdoctoral Associate*
PI: Prof. Lina Necib

Education

2021–2025 **University of Cambridge (Gonville & Caius College)**, *Ph.D. Astronomy & Astrophysics*
Thesis: *Chemodynamics of the Galactic Halo*
Supervisors: Prof. Vasily Belokurov & Prof. N. Wyn Evans

2020–2021 **University of Edinburgh**, *MSc (with Merit) Theoretical Physics*
Thesis: *Probing the Neutrino Mass Hierarchy with Cosmological Density and Velocity Fields*
Supervisor: Prof. John Peacock

2016–2020 **Princeton University**, *A.B. (cum laude) Astrophysics with Applied Mathematics*
Thesis: *Smoothing the Universe by Cosmological Contraction in Various Scalar Field Potentials*
Supervisor: Prof. Paul Steinhardt

Publication Summary

ORCID iD **0000-0001-5996-4072**
H-index **9 (NASA ADS)**
Citations **300+ (NASA ADS)**
Paper count **8 (first author); 9 (other author)**

First Author Publications

Submitted **The erasure of Galactic bar resonances by dark matter subhaloes**, *MNRAS*
Elliot Y. Davies, Adam M. Dillamore, Vasily Belokurov
https://arxiv.org/abs/2603.04490

7 Apr '25 **Blind source separation of the stellar halo**, *MNRAS*
Elliot Y. Davies, Vasily Belokurov, Andrey Kravtsov, Stephanie Monty, GyuChul Myeong, N. Wyn Evans, Sarah G. Kane
https://arxiv.org/abs/2410.21365

22 Feb '24 **Hints of a disrupted binary dwarf galaxy in the Sagittarius stream**, *MNRAS*
Elliot Y. Davies, Stephanie Monty, Vasily Belokurov, Adam M. Dillamore
https://doi.org/10.1093/mnras/stae559

5 Jan '24 **Disrupted dwarf binary merger as the possible origin of NGC 2419 and Sagittarius stream substructure**, *MNRAS*
Elliot Y. Davies, Vasily Belokurov, Stephanie Monty, N. Wyn Evans
https://doi.org/10.1093/mnrasl/slae001

17 Jul '23 **Stellar halo striations from assumptions of axisymmetry**, *MNRAS*
Elliot Y. Davies, Adam M. Dillamore, Vasily Belokurov, N. Wyn Evans
https://doi.org/10.1093/mnras/stad2138

14 Feb '23 **Accelerated phase mixing in the stellar halo due to a rotating bar**, *MNRAS*
Elliot Y. Davies, Adam M. Dillamore, Eugene Vasiliev, Vasily Belokurov
https://doi.org/10.1093/mnrasl/slad017

9 Dec '22 **The phase space chevrons of a GSE-like merger as a dark matter subhalo detector**, *MNRAS*
Elliot Y. Davies, Eugene Vasiliev, Vasily Belokurov, N. Wyn Evans, Adam M. Dillamore
https://doi.org/10.1093/mnras/stac3581

23 Jan '20 **Fuzzy dark matter soliton cores around supermassive black holes**, *MNRAS*
Elliot Y. Davies, Philip Mocz
https://doi.org/10.1093/mnras/staa202

Other Publications

Submitted **Baryon-Accelerated Core Collapse in SIDM Halos and its Imprint on Galactic Disks**
Zeineb Mezghanni, **Elliot Y. Davies**, Adriana Dropulic, Gonzalo Herrera, Abdelaziz Hussein, Lina Necib
https://arxiv.org/abs/2609.05600

Submitted **Galactic Amnesia: The Information Washout of the Milky Way Merger History**
Lina Necib, Dylan Folsom, **Elliot Y. Davies**, Nathaniel Starkman, Andreas Thoyas
https://arxiv.org/abs/2605.04138

Submitted **Formation and Redshift Evolution of Dark Matter Spikes**
Gonzalo Herrera, Abdelaziz Hussein, Lina Necib, **Elliot Y. Davies**, Xuejian Shen
https://arxiv.org/abs/2605.01023

27 Sep '25 **STRRINGS: STReams in Residual Images of Nearby GalaxieS**
Elisabeth Sola, David Chemaly, Vasily Belokurov, Oliver Müller, Anke Ardern-Arentsen, **Elliot Y. Davies** , ...
https://doi.org/10.1093/mnras/staf1647

Submitted **The realm of Aurora. Density distribution of metal-poor giants in the heart of the Galaxy**
Evgeny P. Kurbatov, Vasily Belokurov, Sergey Koposov, Andrey Kravtsov, **Elliot Y. Davies** , ...
https://arxiv.org/abs/2410.22250

8 Aug '24 **The ratio of [Eu/Si] differentiates accreted/in-situ Milky Way stars across metallicities, as indicated by both field stars and globular clusters**
Stephanie Monty, Vasily Belokurov, Jason L. Sanders, Therese T. Hansen, Erika M. Holmbeck, Charli M. Sakari, Madline McKenzie, GyuChul Myeong, **Elliot Y. Davies**, ...
https://doi.org/10.1093/mnras/stae1895

17 Jul '23 **Stellar halo substructure generated by bar resonances**, *MNRAS*
Adam M. Dillamore, Vasily Belokurov, N. Wyn Evans, **Elliot Y. Davies**
https://doi.org/10.1093/mnras/stad2136

25 Nov '22 **Energy wrinkles and phase-space folds of the last major merger**, *MNRAS*
Vasily Belokurov, Eugene Vasiliev, Alis J. Deason, Sergey E. Koposov, Adam M. Dillamore, **Elliot Y. Davies**, ...
https://doi.org/10.1093/mnras/stac3436

13 Aug '20 **Robustness of Slow Contraction to Cosmic Initial Conditions**, *JCAP*
Anna Ijjas, William G. Cook, Frans Pretorius, Paul J. Steinhardt, **Elliot Y. Davies**
https://doi.org/10.1088/1475-7516/2020/08/030

Teaching & Supervision

2022-2024 **Part II Stellar Dynamics and the Structure of Galaxies**, *Supervisor*, Institute of Astronomy, Cambridge

2022 **Part II Introduction to Cosmology**, *Supervisor*, Institute of Astronomy, Cambridge

Mentoring

2026 - 2028 **Daria Zaremba**, *PhD*, MIT, Kavli Institute

2026 **Mark Gaskins**, *Undergraduate (MSRP)*, North Carolina A&T State University

2025 - 2028 **Zeineb Mezghanni**, *PhD*, MIT, Kavli Institute

2026 **Jailee Young**, *Undergraduate (USRP)*, MIT Kavli Institute

2023 - 2024 **Yuxuan Zeng**, *Part III (Masters)*, Institute of Astronomy, Cambridge

Conferences, Talks & Schools

Aug' 26 **Poster entitled The erasure of Galactic bar resonances by dark matter subhaloes**
New Frontier in Cosmology (A Coruna, Spain)

Jun' 26 **Talk entitled The erasure of Galactic bar resonances by dark matter subhaloes**
57th DDA Meeting (Chicago, IL)

Apr' 26 **Talk entitled Detecting dark matter substructure with Milky Way's dynamics tool box**
CIERA Theory group at Northwestern (Evanston, IL)

Apr' 26 **Talk entitled Detecting dark matter substructure with Milky Way's dynamics tool box**
CPC Seminar at Fermilab (Batavia, IL)

Feb' 26 **Talk entitled Detecting dark matter substructure with Milky Way's dynamics tool box**
LPPC Harvard Weekly Seminar (Cambridge, MA)

Jan' 25 **Dissertation talk entitled Blind source separation of the stellar halo**
245th AAS Meeting (National Harbor, MD)

Dec' 24 **Winter school attendee**
Galaxy modelling and Galactic Centre workshop (University of Surrey, Guildford)

Aug' 24 **Contributed talk entitled Hints of a disrupted binary dwarf galaxy in the Sagittarius stream**
Small Galaxies, Large Questions (Durham, UK)

May' 24 **Contributed talk entitled Hints of a disrupted binary dwarf galaxy in the Sagittarius stream**
The Milky Way Assembly Tale (Bologna, Italy)

Sep' 23 **Contributed talk entitled Capturing subhaloes with Galactic halo substructure**
Revealed by Gaia: the central halo of the Milky Way (Institute of Astronomy, Cambridge)

Jul' 23 **Poster entitled Accelerated phase mixing in the stellar halo due to a rotating bar**
European Astronomical Society (Krakow, Poland)

Jul' 23 **Poster entitled Accelerated phase mixing in the stellar halo due to a rotating bar**
Galactic Bars: Driving and Decoding Galaxy Evolution (Granada, Spain)

Mar' 23 **Talk entitled The wrinkles and folds of the Milky Way's most recent major merger**
MCR Research Day 2023 (Gonville & Caius College, Cambridge)

Nov' 22 **Talk entitled The Phase Mixed Debris of a Massive Merger as a Dark Matter Subhalo Detector**
Wednesday Seminar (Institute of Astronomy, Cambridge)

Jan' 20 **Poster entitled How Supermassive Black Holes impact the Fuzzy Dark Matter Mass Mystery**
235th AAS Meeting (Hawai'i Convention Center, Honolulu, HI)

Activities

Nov.'21 – present **Interviewer & Ambassador**, *Princeton University Undergraduate Admissions*
Conduct interviews with prospective Princeton University students and produce reports on each interviewee to aid in the undergraduate admissions process. **Total interviews conducted: 16**

Nov.'21 – Mar.'22 **Journal Club Coordinator**, *Institute of Astronomy, Cambridge*
Organiser for the bi-weekly undergraduate journal club for final year undergraduates and masters students.

Jan.'17– Jan.'18 **Director & Volunteer**, *Princeton Envision (Entrepreneurship Club)*
Assisted in organising, promoting and managing Envision Conference: an annual conference held over 3 days in December that brings 200 competitively selected students from universities across the world together with top researchers, entrepreneurs, and professionals to attend lectures and workshops discussing the disruptive technologies such as artificial intelligence and space travel.

Sep.'17– Mar.'20 **Library Assistant**, *Princeton University Architecture Library*
Duties included but were not limited to: circulation services, reserves, shelving, shelf reading, with other duties as directed by the librarian.

Dec.'16– Sep.'17 **Peer Reviewer**, *Princeton Undergraduate Research Journal*
Assisted in the peer reviewing process for the Princeton Undergraduate Research Journal (PURJ), Princeton's first peer-reviewed academic journal, highlighting top-quality research performed by Princeton undergraduates across all academic pursuits, including the natural sciences, humanities, engineering, social sciences, and arts.

Awards & Academic Programmes

Oct.'21 – Apr.'25 **STFC PhD Studentship**
Cambridge, UK
Full funding for 3.5 years to complete PhD at the Institute of Astronomy, University of Cambridge.

Sep.'16 – Jun'20 **Full-ride Academic Scholarship**
Princeton University, USA
Various financial aid packages covering all tuition fees and living costs to attend Princeton University in the Unites States for four years.

Mar.'15 – May.'16 **Sutton Trust USA Programme**
London, UK
One of 150 selected from 1600 high achieving low-income, first-generation students to receive admissions guidance for applying to the top universities in the US.

Jul.'15 **UNIQ Summer School**
University of Oxford, UK
One of 36 Sixth Form students selected to attended the Oxford University Physics UNIQ Summer School for students from underrepresented backgrounds.